Perturbation Catalogue

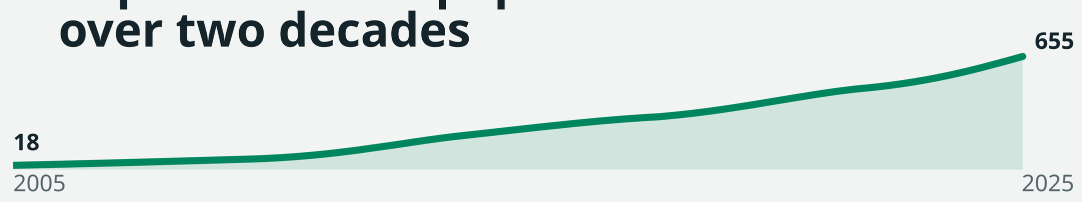
An Integrated Resource for Exploring the Impact of Genetic Perturbations

Kirill Tsukanov, Aleksandr Zakirov, Alexey Sokolov, Mallory Freeberg
European Bioinformatics Institute, Wellcome Genome Campus, Hinxton, United Kingdom

Perturbation evidence is multiplying faster than it can be organised

A single pooled genome-scale Perturb-Seq dataset can **simultaneously profile tens of millions of cells across hundreds of thousands of perturbations**. That scale could reveal robust biology across studies, but identifiers, metadata, assays and processing pipelines differ. The Perturbation Catalogue aims to unify datasets across modalities and provide a shared access path for this evidence.

36× growth
in perturbation papers over two decades



Europe PMC title/abstract query; 24 Aug 2026.
Scale example: FirstWord Pharma, story 7139212.

Our approach turns scattered datasets into a unified resource

1,237	19,235	36	34	1,200	169
datasets	targets	tissues	cell types	cell lines	diseases

CRISPR screens · 1,201 datasets
Perturbed gene → phenotype / fitness changes.

Perturb-Seq · 26 datasets
Perturbed gene → scRNA-seq, cell-wide transcriptome changes.

MAVE · 10 datasets
Combinatorial variant library → sequence-function maps.

We extensively curate metadata...

84 meta-data fields

Study & provenance
Title, authors, year, URI, licence

Perturbation & library
Targets, guides, library, variants

Model system
Organism, tissue, cell line, disease

Readout & processing
Assay, technology, kit, reference

Ensembl gene IDs

Ontology term mapping

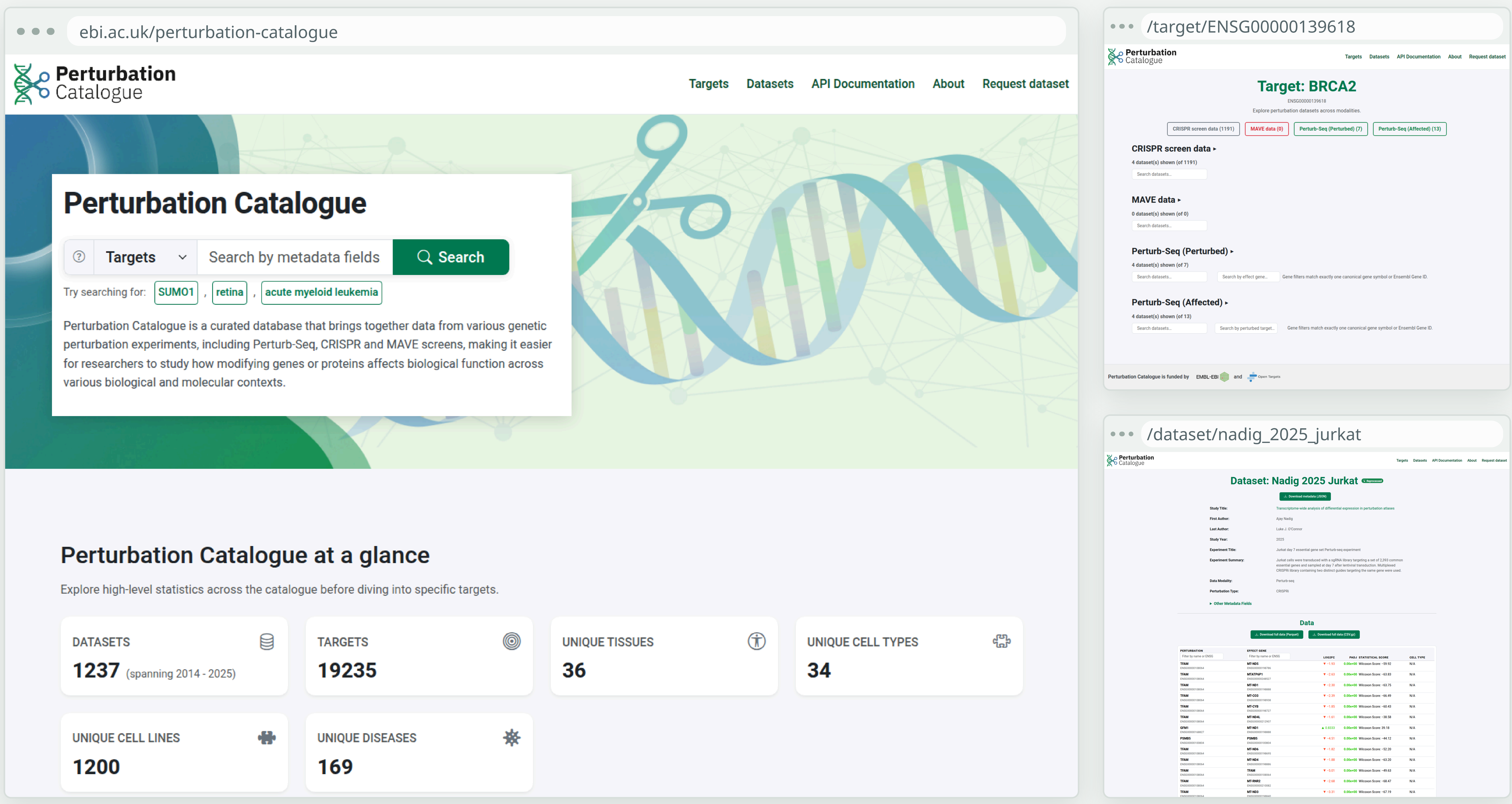
And reprocess data from scratch

SRA / FASTQ — Raw reads + study-specific probes
→ **Cell × gene counts** — Shared reference + gene annotation
→ **Guide assignment** — Map cells to perturbations
→ **QC + filtering** — Final analysis-ready H5AD

Five Perturb-Seq datasets are live. Perturbation-control comparisons feed DEA and Hallmark GSEA. More datasets and modalities coming soon.

Browse the Catalogue

Search → Context → Filter → Explore



Access records via API

Structured JSON for custom pipelines, reproducible analysis and machine learning.

```
GET /v1/perturb-seq/nadig_2025_jurkat/search?perturbation_gene_name=TFAM&limit=1
```


```
{
  "perturbation": {
    "perturbed_target_ensg": "ENSG00000108064",
    "gene_symbol": "TFAM"
  },
  "effect": {
    "effect_gene_ensg": "ENSG00000210082",
    "gene_symbol": "MT-RNR2",
    "log2fc": -2.682
  }
}
```


Download entire datasets

Data is available in **Parquet** and **CSV.gz**, and metadata as **JSON**.

Stay tuned for more exciting things coming soon

Extract structured metadata faster
AI-assisted drafting with curator review and validated ontology mapping.

Reanalyse more designs
More kits, experimental structures and independent quality metrics.

Publish regular, versioned releases
Explicit provenance and stable snapshots for reproducible downstream use.

Reanalyse MAVE data
Enrich variant-to-function coverage with harmonised scores.

Visit: **ebi.ac.uk/perturbation-catalogue** Contact: **perturbation-catalogue-help@ebi.ac.uk**

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